Assignment 01 Report
Deep Learning for Space, Time, And Graphs

October 5, 2026

Github Repository

1 Forecasting Across Heterogeneous Sensors

1.1 Response 1

1.1.1 Task 1A : Predict, measure, and revise the aggregate ordering.

Expected ranking

1. Period-Routing Ridge
2. Raw attention
3. Shared Ridge
4. Sensor Specific Ridge (except station 4)

Output 1.1 results shown in Table 1 and 2 compare the RMSE and MSE for each station and product: (these comparison numbers have been rounded off but preserve enough information for a fair comparison)

Among both the established station, the best RMSE is reported on Period-routing ridge at 0.166 which lines up with the expected ranking. Period-routing is at least 38% better than the next one which is Raw attention at 0.435. Sensor specific and shared ridge rank in the last with a near identical RMSE of 0.768 and 0.767 both at lease 48% worse than raw attention.

For the held-out station, the top ranking one is still Period-routing ridge (0.173) regardless of the minor increase in RMSE. Compared to the significant rise in error for raw attention which was recorded at 0.662 showing its inability to adequately adapt to unseen data. Finally, Shared ridge has the highest error of 0.876.

For product-wise comparison, we observe that shared ridge performs the worst on Product A and B for both established and held out stations as compared to product C with an average difference of 0.2. The predicted order matches with both the established and held out stations in terms of RMSE. However, if we were to extend the analysis based on the parameter count and time cost, Raw Attention would be revised to be much lower in ranking. Additionally, even though Raw Attention performs better than Shared Ridge on the held-out station, the comparative increase

model	RMSE (m/s²)	MSE (m/s²)²	Product A RMSE	Product B RMSE	Product C RMSE	parameters	fit seconds
Shared ridge	0.767	0.588	0.805	0.854	0.621	4608.000	0.032
Sensor-specific ridge	0.768	0.590	0.804	0.852	0.630	13824.000	0.135
Period-routed ridge	0.166	0.028	0.181	0.164	0.152	59904.000	0.040
Raw Attention	0.435	0.190	0.402	0.480	0.421	368368.000	76.407

Table 1: Performance on established stations.

model	RMSE (m/s²)	MSE (m/s²)²	Product A RMSE	Product B RMSE	Product C RMSE	parameters	fit seconds
Shared ridge	0.876	0.767	0.909	0.982	0.713	4608.000	0.032
Sensor-specific ridge	∼	∼	∼	∼	∼	13824.000	0.135
Period-routed ridge	0.173	0.030	0.184	0.182	0.150	59904.000	0.040
Raw Attention	0.662	0.438	0.693	0.597	0.692	368368.000	76.407

Table 2: Performance on held-out station 4.

in error between each of their performance on established and held-out station shows that Raw Attention is not adapting to unseen data well and would be a poor choice.

Period-routing ridge's performance can be explained by how closely it is learned from the data; it is given the exact period values and captures macro trends well [3] based on the assumption that the data's trends are defined and closely related to its pace.
Raw Attention on the other hand relies heavily on discovering trends from the window making it an unreliable choice for unseen data. Shared ridge, as described in the DL4STG-PA1.pdf, assumes that one linear rule will fit (shared) across all the windows which does not work well with the data considering.
Lastly, sensor-specific ridge is based on the assumption that the pace remains the same per station which does not apply to the generated data thereby failing at performance.
Revised Ranking:

1. Period-Routing Ridge
2. Shared Ridge
3. Raw attention
4. Sensor Specific Ridge

Since the pace changes in the same station, using a single matrix W will learn to work with a smoother signal distribution over an average pace rather than learning the nuances of the variable paced signals. Ideally, there would be multiple W to adapt to the peaks and crests of varying paces similar to adding multiple heads in an attention block to attend to different aspects of the data.
In the forecast plot, this materializes as a smoother (reduced amplitude) forecast boundary compared to the ground truth; the discrepancy is more prominent in Product A where the difference is $\tilde{2}$ m/s^2.
However, Product C faces much less relative degradation but the curve almost flattens out after time = 1 leaving very little information to be inferred from the data.

1.1.2 Task 1B : Explain why the forecasting rule must depend on context.

Since the pace changes in the same station, using a single matrix W in Equation 1 will learn to work with a smoother signal distribution over an average pace rather than learning the nuances of the variable paced signals. Ideally, there would be multiple W to adapt to the peaks and crests of varying paces similar to adding multiple heads in an attention block to attend to different aspects

Station	Product	Actual Aeriod (samples)	Model	Window RMSE (m/s^2)
1	Product A	16.135	Shared ridge	0.779
1	Product A	16.135	Sensor-specific ridge	0.787
1	Product A	16.135	Period-routed ridge	0.232
1	Product A	16.135	Raw Attention	0.266
1	Product C	30.093	Shared ridge	0.448
1	Product C	30.093	Sensor-specific ridge	0.436
1	Product C	30.093	Period-routed ridge	0.150
1	Product C	30.093	Raw Attention	0.176

Table 3: Output 1.2

of the data.

$$\hat{z} = x^\top \hat{W}, \qquad \hat{W} = \left(\frac{X^\top X}{n} + \lambda I \right)^{-1} \frac{X^\top Z}{n} \in \mathbb{R}^{L \times H} \tag{1}$$

In the forecast plot, this materializes as a smoother (reduced amplitude) forecast boundary compared to the ground truth; the discrepancy is more prominent in Product A where the difference is 2 m/s2.

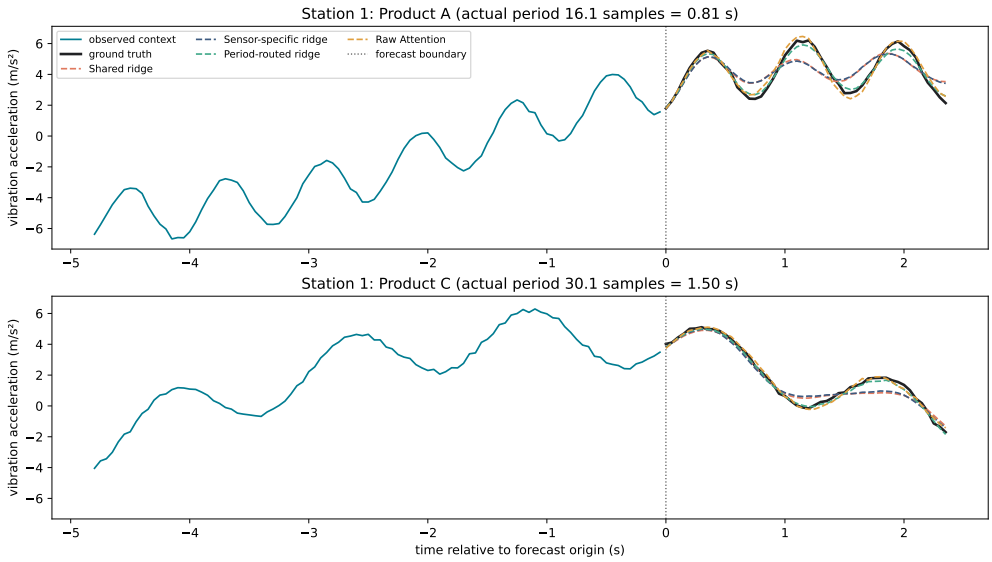


Figure 1: Output 1.2 Plot

However, Product C faces much less relative degradation but the curve almost flattens out after time = 1 leaving very little information to be inferred from the data.

The learned matrices Wv,Wq, and Wk learn from different parts of the distributions; they extract different intricacies of the data. As shown in Figure 2 Product A has more weight given to higher positions (70-100) while C has more of a distributed weighing. The difference in attention concentration allows the matrices to produce different value-mixing operations.

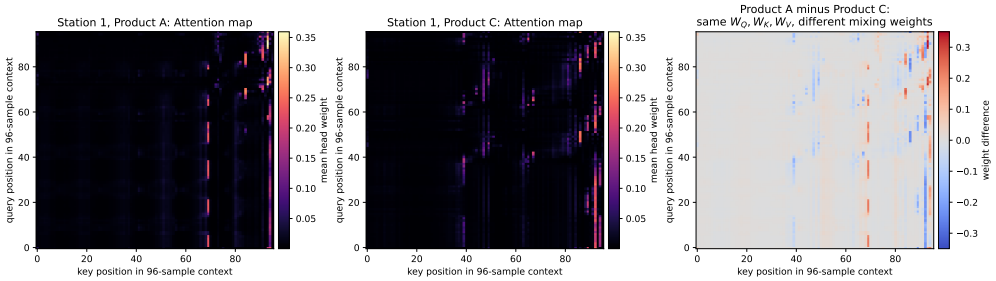


Figure 2: Output 1.3 Attention Map

1.2 Response 2

Width = 41 is chosen since for station 2, the trend is the smoothest among all the values visually. (Figure 3)

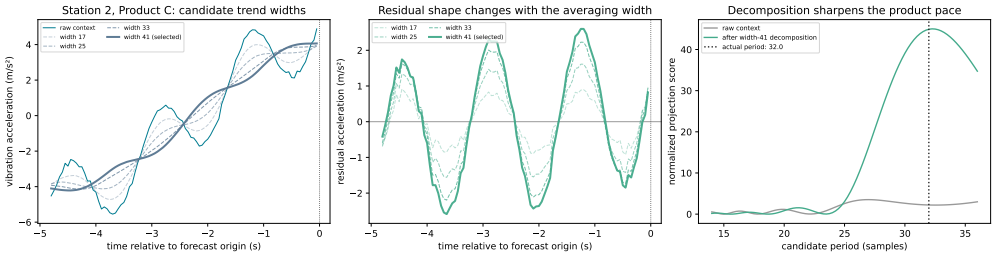


Figure 3: Output 2.1

Empirically, from Table 4 we can observe that the oracle recovery reaches a near perfect value of 99.8% at width = 41 which is highest among all. Continuing with RMSE for validation analysis, for established stations, the highest decrease in RMSE is with Product C of 0.130. For held-out stations, the biggest drop in RMSE is observed in Product A of 0.289.

Representation	Oracle Recovery Within 1 Sample	MdAPE(samples)	Selected
Raw	0.687	0.525	FALSE
Width 17	0.980	0.157	FALSE
Width 25	0.989	0.122	FALSE
Width 33	0.993	0.142	FALSE
Width 41	0.998	0.216	TRUE

Table 4: Output 2.2

Why a moving average is a sensible inductive bias for this drift-plus-repetition world

Moving average acts as a low-pass filter on the 96 sample window by filtering out the high frequency noise which are the repetitive cycles and minimizing group delay when we halve the sample window to 48 while retaining a sharp step response [6] [2]

The drift-plus-repetition attribute of data can cause standard loss metrics to spike when it wrongfully starts to learn patterns in the data leading to overfitting. Oracle diagnostic helps pre-

Station	Product	Expected Delay (samples)	Nearest Integer Delay	Residual Signal Correlation
2	Product A	16.135	16.000	0.896
2	Product A	32.269	32.000	0.897

Table 5: Output 3.2

vent over-inflation of the model to accommodate the drifts and allows for a better selection of width.

Give one situation where this bias would fail or name one alternative decomposition and the different assumption it would encode

When using moving average, a continuous upward or downward drift will cause the model to lag behind the actual trend based on previous records; the inductive bias will fail [4]

1.3 Response 3

The strongest recurrence at P=16.1 and 2P= 32.3 which are near identical to the actual P and 2P for Product A showing a strong positive correlation

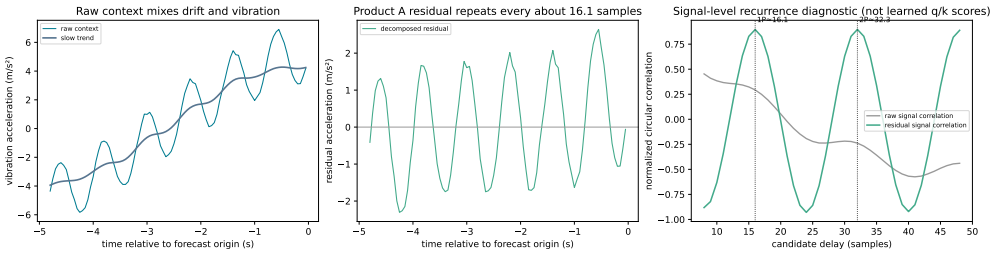


Figure 4: Output 3.2

How decomposition makes the delay evidence cleader than the raw signal which supports the delay mixer's inductive bias

Decomposition eliminates the low-frequency noise and baseline shifts so they do not dominate the variance and prevent isolation of lag between the two signals. Removing them first can prevent spurious correlation. Decomposition also sharpens the cross-correlation function into a sharper peak making it more discernible.

Where delay mixing assumption fails In case there occurs an anomaly in the sensor(s), the cause of an anomaly may not manifest in all the sensors instanly and one sensor may experience a spike in signals while the rest receive the signal later causing the model to try and look for a cross correlation where it might not exist since the anomalous event is an isolated recording which contradicts the recurring displacements assumption [8].

1.4 Response 4

(A)Decomposition would help most on contexts with slow components due to its ability to remove low frequency dominating shifts and preventing spurious correlation.

(B) The tables 6 and 7 report the validation RMSEs and MSEs for the four neural models and the linear baselines on the established stations.

Model	Best RMSE (h)	Worst RMSE (h)	Mean RMSE
Period-routed ridge	**0.107** (2)	**0.229** (46)	**0.162**
Autoformer-inspired	0.134 (1)	0.258 (40)	0.195
Raw delay mixer	0.120 (1)	0.354 (48)	0.212
Attention + decomposition	0.150 (1)	0.423 (48)	0.299
Raw Attention	0.152 (1)	0.557 (39)	0.423

Table 6: Horizon-wise RMSE (m/s^2) on the established stations (1–3). Best and worst of the 48 forecast steps per model, with the horizon step h in parentheses, and the mean over all steps. Models are ordered by mean RMSE.

Model	Best RMSE (h)	Worst RMSE (h)	Mean RMSE
Period-routed ridge	**0.111** (12)	**0.245** (45)	**0.168**
Autoformer-inspired	0.142 (1)	0.340 (45)	0.238
Raw delay mixer	0.138 (1)	0.606 (48)	0.307
Attention + decomposition	0.163 (1)	0.666 (44)	0.440
Raw Attention	0.178 (1)	0.922 (41)	0.635

Table 7: Horizon-wise RMSE (m/s^2) on the held-out Station 4. Best and worst of the 48 forecast steps per model, with the horizon step h in parentheses, and the mean over all steps. Models are ordered by mean RMSE.

Switch	Starting from	Established	Held-out
Decomposition	Attention	−0.094	−0.222
Delay mixing	Attention	−0.141	−0.332
Delay mixing	Decomposed	−0.057	−0.156
Decomposition	Delay mixer	−0.009	−0.046
Both	Attention	−0.150	−0.379

Table 8: Change in error from each architectural switch (from Output 4.2).

According to the results in Table 8, delay mixing helps more than decomposition contrary to the prediction with a reduction of 0.141 in established. However, adding them both further reduces the error showing the importance of both switches (0.150) As the horizon increases, the models perform worse with the least degradation seen in period-ridge routing

(C) Overall, the best model is Period-routing ridge with the consistently low RMSE on both established and held-out stations. the minimal decrease in RMSE of 0.1 is the ideal for the model. Although it has more parameters than Shared ridge and Sensor-specific, it outranks them in performance especially since sensor specific is not applicable on the held-out station at all. It also takes the second lowest time to fit (0.04s) next to shared ridge (0.032) but the significant difference in RMSE where Period Routing is lower by roughly 0.6 on established and 0.6 on held-out makes it the ideal choice.

2 Leaderboard Challenge

As the starting point, I ran the original AutoFormer [9] model on student_train.csv with horizon 168 to get an idea about its performance. Since the original model has around 14m parameters which is not feasible due to compute constraints, i lowered the d to 64 and ended up with 104k parameters and used that as the maximum number for my architecture's. Even on the minuscale AutoFormer a small ¡50k samples dataset caused a poor performance result with a the worst case RMSE of 103.9 for steps 145 to 168.
The final architecture is an encoder-decoder model with a total of $\tilde{4}$3k parameters that has two input linear layer of dimension 48, two encoder layers, followed by 1 decoder layer, 2 fourier series, 8 heads and shared Q,K. For the decoder, the intial step is to put the the original 96-step window through moving average with a kernel size of 25 to initial the decoder on the residual mode with zero-initialisation [1] [10]. I used sinusoidal positional encoding. The original Autoformer basics are implemented for all comparisons of potential models (referring to various AutoFormer-based architectures). Exploring the data showed that the timestamps were not divided into seasons and no information about the cycle was explicitly stated. Therefore, the repeating cycle had to be discovered from the data. First step was to check the raw series which lead to the discovery that the autocorrelation decays quickly as expected of short range dependent data. It also meant that a simple FFT would not give me the insight that i needed so i chose to add a Welch spectrum analysis [7] (Figure 5) which led to the Fourier value being set at 2 since only periods 24 and 12 were above reference for a lengthier cycle. [11] [5]

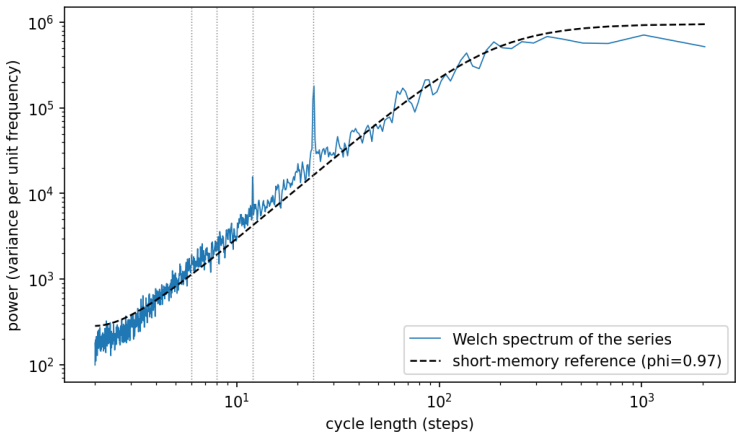


Figure 5: Welch Spectrum

From the ablation studies recorded in Table 10, I picked the best RMSE that was 68 using the model and increased the parameters to check whether the RMSE improved. from the original 1584 to 5045, the RMSE improved to 66 and a further increase to 43k also improved to 50. To finalise, i ran another test with total of 66k parameters but the RMSE maintained at 50 showing that adding any more parameters is not contributing to the learning. The Leaderboard Scores are reported in Table 9 and the forecast plot (Figure 6) shows the 168 next prediction map.

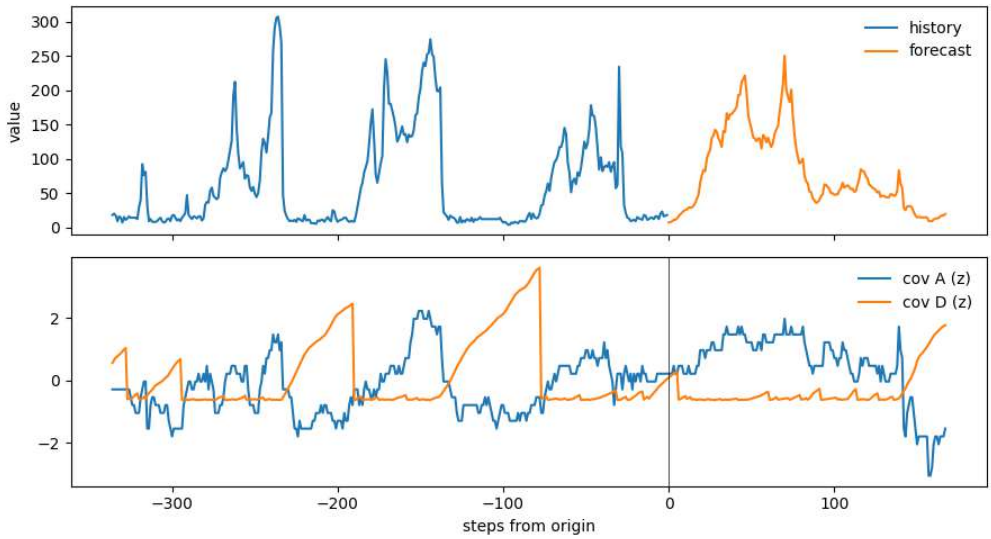


Figure 6: Final Forecast plot

MAE	RMSE	sMAPE	Params	Epochs
55.9806	92.0024	53.25%	42,784	20

Table 9: Leaderboard results

2.1 Ablation

arm	P	RMSE	MSE	MAE	RMSE_sd
baseline: global mean	0	94.363	8904.351	70.677	NaN
baseline: ACF shrink	0	88.905	7904.122	66.283	NaN
plain	596	106.388	11319.719	77.945	1.329
base	599	88.894	7902.199	66.313	0.142
dec+f2	703	72.711	5297.403	53.264	3.613
dec+f2+slow	704	72.577	5276.127	53.032	3.307
dec+f2 untied	1584	68.902	4748.688	50.243	1.256
dec+f2 d16	2415	72.685	5299.526	53.755	4.528

Table 10: Ablation

For the ablation study, since there were multiple runs, I had to restrain the parameters to very low so all of them could be executed within the available compute units. I acknowledge that this introduces a certain uncertainty factor since it is possible that a version is performing poorly because the parameters are too little. It can be further experimented and certainly improved.

References

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AI use: Claude , Gemini inline generator for Colab